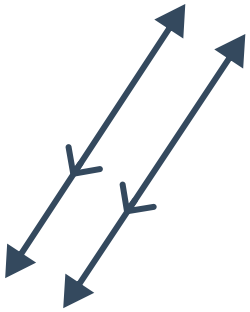
Parallel lines and transversals

1

a Parallel lines are indicated using arrows or 'chevrons'. The number of arrows connects which lines are parallel.

b Example answer:



2

a Corresponding angles **b** Alternate angles

c Cointerior angles

3

a Equal **b** Supplementary **c** Equal

4

a

i $\angle EPQ$ **ii** $\angle APQ$ **iii** $\angle APF$

b

i $\angle CQP$ **ii** $\angle EQP$ **iii** $\angle DQE$

5

a The diagram contains parallel lines. The marked angles are supplementary and cointerior so we have a transversal of parallel lines.

b The diagram does not contain parallel lines. The marked angles are not supplementary so they are not cointerior. We therefore do not have a transversal of parallel lines.

c The diagram contains parallel lines. The marked angles are equal and corresponding so we have a transversal of parallel lines.

d The diagram does not contain parallel lines. The marked angles are not equal so they cannot be corresponding. We therefore do not have a transversal of parallel lines.

e The diagram contains parallel lines. The marked angles are equal and alternating so we have a transversal of parallel lines.

f The diagram contains parallel lines. The marked angles are not equal so they cannot be alternating. We therefore do not have a transversal of parallel lines.

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g The diagram does not contain parallel lines. The marked angles are not equal so they cannot be corresponding. We therefore do not have a transversal of parallel lines.

h The diagram contains parallel lines. The marked angles are supplementary and cointerior so we have a transversal of parallel lines.

6

a $x = 125$ **b** $x = 75$ **c** $x = 55$ **d** $x = 52$

e $x = 107$ **f** $x = 96$

7

a They are alternate angles, so they are also equal.

b They form a straight angle, so they are also supplementary.

c 96°

8

a They are vertically opposite angles, so they are also equal.

b They are cointerior angles, so they are also supplementary.

c 73°

9

a They are cointerior angles, so they are also supplementary.

b They form a straight angle, so they are also supplementary.

c 97°

10

a $x = 33$ **b** $y = 57$

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